Engr. Muhammad Umer Haroon

# Assignment 1

**SOFTWARE CONSTRUCTION AND DEVELOPMENT**

**“Project Initiation, Git Collaboration, and Requirements Documentation”**

**Objective:** To initiate your semester project using collaborative software development practices, establish a shared Git repository, and document the initial requirements of your project using an agile approach.

# Tasks

## 1. Create a Shared Git Repository

- Create a public repository on GitHub for your semester project.
- Each group must consist of **multiple members**. Exceptionally one, who has taken permission.
- Add all group members as collaborators and ensure that everyone contributes to the same repository.
- Each member must use their own GitHub account and make meaningful contributions through commits.
- Maintain a complete and traceable record of project development using Git commits.
- Use meaningful commit messages that clearly describe the changes made.

## 2. Document Project Requirements

Prepare a lightweight requirements document for your project. A formal Software Requirements Specification (SRS) is not required.

The document should include:

- Project title and group member details.
- A brief description of the problem being addressed.
- Project objectives and intended users.
- Initial functional requirements, describing what the application should do.
- Relevant non-functional requirements, where applicable.
- A prioritized list of features or a simple product backlog.
- Initial assumptions, constraints, and known limitations.

Follow an agile approach: document the requirements clearly enough to guide development, while recognizing that they may evolve as the project progresses.

### 3. Organize and Maintain the Repository

- Include the requirements document in the repository.
- Use a clear and consistent folder structure.
- Include a `README.md` file containing the project overview, group members, and current project status.
- Establish and follow a consistent naming convention for files, folders, and project versions.
- Use Git commits and tags to maintain a history of changes and significant project versions.

For project versions, you may use a convention such as `v0.1.0` for an initial development version and `v1.0.0` for the first complete release. Do not create separate copies of files merely to represent different versions; use Git history and tags instead.

### 4. Work as an Agile Team

- Divide initial responsibilities among group members.
- Maintain a simple task list or product backlog.
- Ensure that all group members participate in project development and repository activities.
- Update the requirements and task list as the project evolves.

## Submission Requirements

Submit the following:

1. **Public GitHub repository link** — the repository must be accessible without requesting permission.
2. **Requirements document** — submitted as a PDF or other instructor-approved format.
3. **Repository copy of the requirements document** — the same document must also be committed to the repository.

Only **one submission per group** is required. One designated group member may submit the repository link and requirements document on behalf of the entire group.

However, all group members are individually responsible for contributing to and understanding the work in the shared repository.

**Important:** This assignment establishes the foundation for subsequent project milestones. Your repository, requirements, and development history should be maintained throughout the semester rather than created only for submission. Copied / plagiarized, massively AI generated work, (identical working or identical mistakes) will receive zero for everyone involved. Random students may be asked to explain their solutions in a short viva.